

VARREN HANSEN

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EXPERIENCE

Deloitte Advisory | *Analyst – Forensics & Financial Crime*

Sep 2025 – Present

- Leading the development of two web-based solutions addressing KYC operational pain-points for a global financial institution.
- Conducting requirements analysis and translating business specifications into scalable technical solutions for diverse stakeholders.
- Architected reusable React components adhering to enterprise UI/UX standards, ensuring seamless integration for complex workflows.
- Integrated PowerBI dashboards into case management platforms using iframes and led a backend team to develop APIs writing to SharePoint.
- Serving as lead UI developer, triaging bugs and resolving issues across 5 production environments to support end-users.
- Strategic code refactoring and bundle optimization to ensure scalability for enterprise-scale operations.

Deloitte Advisory | *Co-op Intern – Forensics & Financial Crime*

Jun 2024 – Jan 2025

- Architected a news screening platform featuring RAG-based semantic search and real-time model evaluation visualization.
- Optimized SQL queries and database schemas, maintaining support for 100 concurrent users during peak operations validated via AWS Lambda.
- Established an automated test suite achieving 90% code coverage, enabling deployments with zero downtime.

DocPro Limited | *Front-end Developer (Intern)*

Jun 2023 – Aug 2023

- Developed the front-end interface for an AI-powered chat feature using React and JavaScript.
- Implemented prompt engineering techniques to enhance the accuracy and performance of the company's AI models.
- Delivered high-quality results within a Scrum Agile environment through iterative development.

PROJECTS

Porkfolio | *Expense Tracker Mobile App*

May 2025

- Built a full-stack MERN (MongoDB, Express, React Native, Node.js) mobile application for financial data visualization and spending analysis.
- Implemented secure user authentication using JWT and encrypted password handling.
- Demo: <https://youtu.be/CzXCHZGC6ak>

Bitcoin Price Trend Analysis & Prediction | *Course Related Project*

May 2025

- Leveraged PySpark and Random Forest to process 2.2M+ records, engineering a dataset with an 11-day lookback window to achieve a 0.999 R² prediction accuracy.

Document Gap Analysis | *Co-op Project - Deloitte Advisory*

Nov 2024 – Dec 2024

- Developed a full-stack NextJS application leveraging Retrieval-Augmented Generation (RAG) to identify compliance gaps in regulatory documents.
- Automated document analysis processes by combining GenAI with user input, providing actionable compliance insights.

EDUCATION

The Hong Kong University of Science and Technology | *BEng in Computer Science*

Sep 2021 – May 2025

- Dean's List – Spring 2024/25

SKILLS

- **Web Development:** React, React Native, Vue.js, Angular, NextJS, HTML, CSS, JavaScript, TypeScript, Figma.
- **Server-Side & Databases:** Node.js, Python, Java, C++, SQL, PostgreSQL, MongoDB, RESTful API design.
- **AI & Data:** Generative AI, Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, PowerBI.
- **Cloud & DevOps:** AWS (EC2, Lambda), SharePoint integration, Automated Testing (Mocha, Chai, Selenium).
- **Languages:** English (IELTS 7.5), Indonesian (Native), Chinese (Limited Proficiency).